Hot-stove learning sustains masking when authenticity is optimal: reinforcement-learning models of gifted peer adaptation

Original Research

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Abstract

Persistent learned masking emerged even when authenticity was the objectively optimal policy, and this suboptimal regime occupied most of the maintained M2 phase map. We developed five reinforcement-learning models in which masking was safe and authenticity exposed the learner to uncertain peer response. The models isolated risky authenticity, graded disclosure, stigma states, multidimensional context search, comparison cost, and endogenous receptivity. On the coarse M2 grid, 226 of 357 cells (63.3%) were classified as learned suboptimal masking: value iteration favored authenticity in the neutral state while the learned neutral action remained masking. Across these cells, the mean learned neutral-state masking probability was 0.950. At $p = 0.55$, it was 0.984 without stigma escape and 0.973 with escape $\eta = 0.20$, despite authenticity being optimal in both environments. This is the central hot-stove result: negative experience suppresses further sampling of authenticity and allows masking to persist without being the best available policy. Stronger negative-outcome weighting increased masking by 0.403, while higher receptivity increased graded disclosure by 0.569. Multidimensional search compounded the burden: increasing contexts reduced discovery of the receptive context in flat and hierarchical action spaces, but not when masking was global or the good context was known. A large comparison cost eliminated good-context authenticity in the tested condition. Endogenous receptivity made masking self-reinforcing and produced distinct ascending and descending paths. The simulations therefore identify several routes by which persistent masking can be learned; their prevalence in human populations remains an empirical question.

1 Introduction

Students identified as gifted do not form a psychologically uniform group. Large-sample and systematic-review evidence does not support a stable profile of generalized social or mental-health difficulty that separates all gifted young people from their peers (Lavrijsen and Verschueren, 2023; Tourreix et al., 2023). At the same time, a distinct literature documents situations in which gifted students manage what peers know about their ability. Early work on the stigma of giftedness described a tension between being recognized as different and maintaining ordinary social interaction (Coleman and Cross, 1988; Cross et al., 1993). Later measurement and interview studies identified strategies such as denying giftedness, conforming, using humor, helping peers, or otherwise reducing

the visibility of exceptional performance (Swiatek, 1995; Rudasill et al., 2007; Cross et al., 2019). These findings motivate a contextual question: under what social contingencies could reduced visibility become a learned and persistent response?

This question is educationally important because the immediate and delayed consequences of disclosure can diverge. Showing ability, intensity, or unusual interests may be intrinsically valuable and may eventually support compatible friendships, yet a rejected attempt can be salient. Masking may carry an ongoing cost, but it can also deliver a comparatively predictable short-term return. Sequential experience therefore creates an information problem. A learner who receives an early negative outcome from the risky action samples that action less often and obtains less evidence about its long-run value. This hot-stove mechanism can produce apparent risk aversion without assuming a fixed risk-averse trait (March, 1996; Denrell and March, 2001). Reinforcement-learning models offer a transparent way to formalize this possibility, while keeping separate the objective environment, the learner's update rule, and the claims that the simulation can support (Sutton and Barto, 2018).

The present study develops a family of five deliberately simple models. M0 is a two-action bandit. M1 replaces the binary action with graded disclosure. M2 introduces neutral, accepted-as-self, and stigmatized states and evaluates the learned policy against an optimal Markov decision process. M3 allows the learner to search among peer contexts and tests whether a context-count effect survives changes to the action representation. M4 allows receptivity itself to change after masking, acceptance, or rejection. These models are related conceptually, but they are not estimated stages of one empirical developmental trajectory. Each isolates a candidate mechanism.

We tested a strong proposition: authenticity can be the best available long-run policy while an experiential learner nevertheless settles into persistent masking. We expected hot-stove information restriction to generate this failure, negative-outcome weighting and sticky stigma to deepen it, graded disclosure to preserve safer routes of exploration, and multidimensional context search to enlarge the burden of finding a receptive setting. We further tested whether masking could reduce future receptivity and thereby become self-reinforcing. Optimal-policy benchmarks and action-space ablations were used to identify when persistent masking reflected the learner's history rather than the objective superiority of masking.

2 Materials and Methods

2.1 Study design

We conducted computational experiments with synthetic learners to isolate how reward risk, learning, state transitions, context search, and behavior-environment feedback affect masking. The action called masking represents selection of the safer, less visibly authentic option under the specified payoff structure. Parameters were chosen to expose mechanisms and edge cases rather than fitted to a human dataset. No human or animal participants, personal data, or clinical records were used.

2.2 Shared learner

All models used independent action values initialized to q0 and updated with learning rate alpha. Bandit models used the incremental update Q(a) <- Q(a) + alpha[v(R) - Q(a)]. M2 used the Q-learning target v(R) + gamma max_a' Q(s',a'), where gamma is the discount factor (Watkins and Dayan, 1992). The subjective transform affected learning only: v(R) = R when R >= 0 and v(R) = lambda R when R < 0. Thus, lambda represents the weight placed on a realized negative outcome; it

does not alter the objective payoff generated by the environment. The maintained implementation permits lambda >= 0 and rejects negative values. Default lambda was 1.

Actions were sampled with a temperature-controlled softmax policy, Pr(a) = exp[(Q(a) - max Q)/tau] / sum_j exp[(Q(j) - max Q)/tau]. Subtracting the row maximum makes the calculation numerically stable. Each parameter cell created a local NumPy random generator from a recorded seed; no model used global random state. The default learning settings were alpha = 0.10, tau = 0.25, and q0 = 0.

2.3 M0: risky authenticity

M0 offered masking and authenticity. Masking produced the deterministic objective reward E(mask) = b - c. Authenticity was accepted with probability p and produced b + a; rejection produced a - r. Its expected objective reward was E(auth) = p(b + a) + (1 - p)(a - r) = a + pb - (1 - p)r. Equating the two yields r*(p) = (a + c)/(1 - p) - b, for p != 1. With b = 1.0, a = 0.6, c = 0.25, and r = 2.0, masking returned 0.75, authenticity returned -1.4 + 3p, and the objective indifference point was p* = 0.7167. We verified the endpoint cases p = 0 and p = 1, stable softmax probabilities, and the separation of objective and subjective rewards in unit tests.

2.4 M1: graded disclosure

M1 used five evenly spaced disclosure levels d from 0 to 1, including both endpoints. Acceptance probability was 1 - (1 - p)d. The objective expected reward was E(d) = a d - c(1 - d) + b[1 - (1 - p)d] - r[(1 - p)d] = b - c + d[a + c - (1 - p)(b + r)]. Therefore d = 0 reproduces masking and d = 1 reproduces full authenticity. Because expected reward is linear in d, an intermediate level is generally not the static optimum except at indifference. Each level had an independent Q-value, so this version tests whether lower-exposure alternatives change exploration; it does not model smooth generalization between nearby disclosure levels.

2.5 M2: stigma-state Markov decision process

M2 used a state-by-action Q-table with three states: neutral, accepted-as-self, and stigmatized. In neutral, masking returned b - c and left the state neutral; authentic acceptance occurred with probability p, returned b + a, and moved to accepted; authentic rejection returned a - r and moved to stigmatized. In accepted, authenticity reliably returned b + a and masking returned b - c. Independent turnover with probability delta then reset the next state to neutral. In stigmatized, masking returned b2 - c and authenticity redeemed the learner with probability p/2, returning b + a and moving to accepted; failure returned a - r2. If the next state remained stigmatized, escape with probability eta reset it to neutral. Rewards were generated from the current state before turnover or escape affected the next state. Masking therefore did not automatically erase stigma.

For selected cells and parameter grids, value iteration solved the objective-reward Markov decision process to a Bellman residual below 1e-10. We compared the optimal neutral-state action with both the static M0 optimum and the learner's final-window empirical policy. The default M2 environment used b2 = 0.5, r2 = 2.5, delta = 0.02, and gamma = 0.9. The principal learner contrast used p = 0.55, eta in {0, 0.20}, 200 learners per cell, and 1,200 steps. A separate threshold analysis used value iteration over eta from 0 to 0.80 and a coarse learner map over p and eta.

2.6 M3: peer-context search and comparison cost

M3 contained K contexts. One good context had receptivity p_good = 0.85 and the remaining contexts had p_bad = 0.20. In the original flat representation, each context contributed a masking and an authenticity arm, giving 2K arms; action 2g represented masking in group g and 2g + 1 represented authenticity. Masking returned b - c in every context. Authenticity used the selected context's p. A comparison cost bf was subtracted only from accepted authenticity in the good context. The objective crossover at which good-context authenticity equaled masking was bf* = {p_good(b + a) + (1 - p_good)(a - r) - (b - c)}/p_good = 0.4706 at the defaults.

To determine whether increasing K reflected context multiplicity or a representational search burden, we compared four formulations: the flat 2K-arm model; a hierarchical model that chose a context and then behavior within that context; a global-mask model with one shared masking action and K context-specific authenticity actions; and a fixed-good-context model that removed context search. Each ablation compared K = 1 with K = 8 using 220 learners per condition for 1,200 steps. The outcome was the proportion whose modal final-window action was authenticity in the good context. Random good-context identities and multiple good contexts were not included in the present artifact and remain planned tests.

2.7 M4: endogenous receptivity

M4 retained two action values but allowed each learner's receptivity p_t to evolve between p_min = 0.05 and p_max = 0.95. Reward was sampled from the current p_t and the Q-value was updated first. Accepted authenticity then increased receptivity by ka(p_max - p_t); rejected authenticity decreased it by kr(p_t - p_min); and masking decreased it by km(p_t - p_min). Values were clipped to the stated bounds after every update. Because Q-values did not condition on p_t, this is an endogenous nonstationary bandit, not stateful Q-learning.

The primary M4 contrast used p0 = 0.45, ka = 0.12, kr = 0.04, km in {0, 0.003}, 220 learners per cell, and 1,200 steps. For the path experiment, km increased over seven evenly spaced values from 0 to 0.003 and then decreased over the same grid. At each value, 220 learners continued from the current Q-values and p_t for 500 steps. The descending path therefore inherited the endpoint reached on the ascending path, as required for a ramp test. It remained a single schedule with one seed sequence.

2.8 Outcomes and statistical summaries

Primary continuous outcomes were the proportion of masking actions or mean disclosure during the final 20% of steps. M3 settling was defined by the modal action in that window. M2 diagnostics also stored state-conditional action counts, objective rewards, and final states. For child-level contrasts, we recomputed the difference between independent condition means and used 10,000 nonparametric bootstrap draws within each condition (seed 90517) to obtain descriptive 95% Monte Carlo intervals. This corrects the repository's earlier administrative pairing of child indices across independently seeded conditions. The intervals quantify within-run simulation dispersion, not uncertainty across master seeds or a human population. The M4 path gaps are descriptive averages across seven matched ramp points; their points are serially dependent, so no population-level inference is attached to them.

2.9 Reproducibility and software

The maintained Python package separates core numerical functions, model kernels, experiment generation, optimal-policy analysis, and reproducibility utilities. The robustness manifest records parameter cells, condition seeds, NumPy generator strategy, horizons, final-window fractions, output

paths, Python and package versions, and a code-configuration hash. Primary unit-level rows are stored in CSV format. Repository verification runs deterministic tests and regenerates the maintained artifacts. The analyzed run used master seed 0 for experiment orchestration and explicit cell seeds 11-24, 301, 308, and 500-513; the bootstrap reported here used seed 90517.

OpenAI Codex (GPT-5; OpenAI, https://openai.com/; accessed 10 August 2026) assisted with manuscript restructuring, editing, reference checking, figure scripting, and conformance checks. It did not generate simulation observations or alter the model kernels. The authors reviewed the code-grounded claims and remain responsible for the manuscript. This disclosure should be reconfirmed by the authors before submission.

3 Results

3.1 Objective benchmarks made suboptimal learned masking identifiable

The analytic benchmarks established where masking was and was not objectively justified. Under the default M0 payoffs, authenticity exceeded masking above $p = 0.7167$. M1 reproduced the same endpoints, and its linear expected-reward function showed that intermediate disclosure changed the path of exploration rather than creating a new static optimum. In M3, good-context authenticity exceeded masking below the comparison-cost crossover $bf = 0.4706$. These thresholds allowed persistent learned masking to be identified as a failure to discover or maintain the higher-value policy, rather than merely as adaptation to an unfavorable environment.

3.2 Risky authenticity and negative-outcome weighting produced persistent masking

Risk alone was sufficient to sustain sharply different learned behavior. In M0, mean final-window masking was 0.995 at $p = 0.20$ and 0.262 at $p = 0.85$, a difference of 0.733 (independent-bootstrap 95% interval 0.721 to 0.745; 250 learners per condition). At $p = 0.65$, where the objective environment was held fixed, increasing negative-outcome weighting lambda from 0.50 to 2.50 raised masking from 0.586 to 0.990, a difference of 0.403 (0.386 to 0.421; 250 per condition). Weighting negative experience more heavily therefore magnified avoidance without changing the objective payoff of authenticity.

3.3 Graded disclosure preserved safer exploration

In M1, mean final-window disclosure was 0.048 at $p = 0.20$ and 0.617 at $p = 0.85$. Higher receptivity increased disclosure by 0.569 (0.556 to 0.582; 250 per condition). The intermediate actions did not become new static optima; they created lower-exposure opportunities to continue sampling disclosure. Graded disclosure therefore preserved exploration that a binary mask-or-authenticate choice could terminate after a negative outcome. Figure 1 and Table 3 summarize the maintained directional contrasts.

3.4 Hot-stove learning produced a large regime of suboptimal persistent masking

At $p = 0.55$, authenticity was the optimal neutral-state action both without stigma escape and with escape $eta = 0.20$. The learners nevertheless masked with neutral-state probabilities of 0.984 and 0.973, producing objective regrets of 3.472 and 3.575. Raising eta reduced mean final-window masking from 0.785 to 0.584, a difference of 0.201 (0.115 to 0.287; 200 learners per condition), but it did not eliminate the learned preference for masking. Sticky stigma therefore intensified a learning failure that already existed: the learner repeatedly selected the safe action even though value iteration showed that authenticity was better.

The failure regime was large, not marginal. On the maintained 21 by 17 grid of receptivity p and escape eta, 226 of 357 cells (63.3%) were classified as learned suboptimal masking. Across those cells, the mean learned neutral-state masking probability was 0.950, with classified cells extending from p = 0.40 to 0.95 at low eta and from p = 0.30 to 0.95 at eta >= 0.40. Meanwhile, the refined dynamic authenticity threshold fell from 0.3839 at eta = 0 to 0.2690 at eta = 0.80, far below the static M0 threshold of 0.7167 (Figure 2). Sensitivity checks showed the same mechanism: at p = 0.75 and eta = 0, lowering exploration temperature increased neutral-state masking from 0.950 to 0.994, whereas higher exploration reduced it to 0.649. A separate environment sweep found only a narrow region in which masking itself was dynamically optimal above the static threshold; the largest tested width was 0.0563 under the harsh setting r2 = 8, delta = 0.30, eta = 0, and gamma = 0.9. The dominant result under the default environment was therefore persistent learned masking despite optimal authenticity.

3.5 Multidimensional context search enlarged the burden of finding authenticity

Adding contexts impaired discovery when it also expanded the action space. In the flat 2K-arm model, the proportion settling on authenticity in the receptive context fell from 0.964 at K = 1 to 0.764 at K = 8, a gap of 0.200 (0.141 to 0.264; 220 per condition). In the dedicated ablation, the K = 1 minus K = 8 gap was 0.168 (0.114 to 0.227) in the flat representation and 0.377 (0.309 to 0.445) in the hierarchical representation. The gap collapsed to 0.005 (-0.027 to 0.036) with a global masking action and -0.005 (-0.032 to 0.023) when the good context was fixed and known (Figure 3). Thus, multidimensional search made the receptive option harder to discover; context count alone did not produce the effect.

3.6 Large comparison costs eliminated good-context authenticity

At K = 4, good-context authenticity was the modal final-window action for 0.918 of learners when bf = 0 and for none when bf = 0.80, a contrast of 0.918 (0.882 to 0.955; 220 per condition). The tested high-cost condition lies above the analytic crossover bf = 0.4706. Once the comparison cost exceeded the objective advantage of the receptive context, the model abandoned that context completely. This establishes comparison cost as a powerful potential counterweight to the benefit of a welcoming peer setting within the model.

3.7 Endogenous receptivity made masking self-reinforcing and path-dependent

When masking reduced future receptivity, it became self-reinforcing. At p0 = 0.45, increasing km from 0 to 0.003 raised final-window masking from 0.304 to 0.712, a difference of 0.409 (0.338 to 0.478; 220 per condition). In the continuation ramp, the mean descending-minus-ascending mask-rate gap across the seven matched km values was 0.065, while the corresponding receptivity gap was -0.106. The paths remained separated after reversal because the ascending path had already changed both learned values and receptivity (Figure 4). The model therefore retained a behavioral history: the same current km produced different outcomes depending on the route by which it was reached.

4 Discussion

4.1 Principal findings

Risky authenticity produced persistent learned masking. This is the central result. Masking persisted even where authenticity was the objectively superior policy, and the effect occupied 226 of 357 cells (63.3%) in the maintained M2 grid. Across those cells, the learned neutral-state masking

probability averaged 0.950. The phenomenon is therefore not confined to a knife-edge boundary: under the tested learning rule, large parts of the authenticity-optimal environment were behaviorally occupied by masking.

Hot-stove persistence was the critical learning mechanism. A negative authentic outcome lowered its learned value, reduced subsequent sampling, and restricted access to corrective positive evidence. Lower exploration intensified the effect, while greater exploration weakened it. Negative-outcome weighting magnified masking without changing objective rewards, and sticky stigma prolonged the consequences of rejection. Persistent masking therefore arose from the interaction between costly experience and information restriction, not simply from a stable preference for concealment.

Graded disclosure preserved a route back into exploration. Intermediate actions were not generally the highest-value endpoints, but they reduced exposure during sampling and allowed disclosure to continue where a binary choice more readily collapsed into masking. The model therefore separates the value of partial disclosure as an exploratory bridge from its value as a final policy.

Multidimensional search enlarged the burden. Adding contexts impaired discovery only when the learner had to search across context-specific alternatives; the effect disappeared when the receptive context was known or masking was represented globally. The result identifies environmental legibility and search architecture, rather than social variety itself, as the active mechanism.

Comparison costs and endogenous feedback can overpower receptivity. A sufficiently large comparison cost eliminated good-context authenticity, while mask-driven receptivity loss more than doubled masking and produced distinct continuation paths. Welcoming conditions can therefore fail to translate into visible authenticity when the learner pays a strong comparison cost or when previous masking has already degraded future returns.

4.2 Implications for educational psychology

The educational implication is direct: persistent masking need not reveal a stable preference for concealment. It can be learned from a sequence in which visible authenticity is occasionally punished, masking is reliably safe, and later positive evidence becomes difficult to obtain. A student may therefore continue masking after the environment has become favorable, or may fail to discover a receptive context that already exists. This sequential account extends the stigma-of-giftedness literature by explaining how an initially adaptive response can outlast the conditions that produced it (Coleman and Cross, 1988; Cross et al., 1993; Denrell and March, 2001).

M1 identifies graded participation as a mechanism for preserving exploration. Lower-exposure actions can keep the learner in contact with informative outcomes after full disclosure has become too risky to sample. The practical hypothesis is not that partial disclosure is always preferable, but that low-stakes opportunities may reopen a path toward fuller participation. Empirical work should test whether safety learned through one form of participation generalizes to nearby actions and whether repeated low-cost successes reverse established masking.

M3 shows that the legibility of the social environment matters. A receptive peer context provides little benefit if it is difficult to identify among many context-specific alternatives while masking remains safe everywhere. Clear signals of receptivity, transferable knowledge about safety, and trusted guides should reduce this search burden. The model predicts that increasing the number of

social options will help only when learners can recognize or efficiently test which options are receptive.

The comparison-cost result adds an important counterforce. Even a highly receptive context ceased to attract authentic behavior once the imposed comparison cost exceeded its reward advantage. This creates a precise theoretical possibility: a setting can be socially welcoming yet still discourage authenticity because belonging there carries a sufficiently high self-evaluative cost. Academic self-concept research makes this mechanism plausible (Marsh and Hau, 2003; Fang et al., 2018), although the model's bf parameter remains uncalibrated to human data.

4.3 Boundary conditions and limitations

First, no human data were fitted or predicted. Terms such as authenticity, stigma, receptivity, and masking are operational labels for model components. Their psychological validity, measurement invariance, and developmental meaning remain untested. The literature also does not support portraying gifted young people as uniformly distressed or socially deficient (Lavrijsen and Verschueren, 2023; Tourreix et al., 2023). The simulations model one possible response to particular contingencies, not a population profile.

Second, the principal Monte Carlo contrasts used one deterministic seed per condition and moderate populations. The independent bootstrap intervals summarize dispersion among synthetic learners within those runs. They do not substitute for replication across independent master seeds, and synthetic learners within a vectorized run share the same parameter cell and temporal schedule. Claims about phase boundaries require denser grids and uncertainty in threshold locations.

Third, constant alpha and tau preserve ongoing learning and exploration. High final-window masking is therefore persistent behavior over the observed horizon, not a mathematically absorbing state. Results may change with decaying learning rates, alternative exploration rules, longer horizons, correlated social outcomes, memory, or model-based inference. The learner parameters have not been mapped to temperament, diagnosis, or stable individual differences.

Fourth, M2's default environment made acceptance valuable enough that dynamic authenticity became optimal below the static threshold. True optimal masking above the static threshold appeared only in a harsher tested regime. Broader sensitivity over gamma, turnover, stigmatized payoffs, eta, alpha, tau, q0, and horizon is required before structural-lock-in regions can be treated as robust. The current learner map also needs independent-seed replication and threshold-local refinement.

Fifth, M3 still lacks random good-context identity by learner and multiple-good-context conditions. M4 lacks ramp-speed sensitivity, independent ramp seeds, loop-area estimates, transition-threshold separation, and controls that distinguish deterministic continuation from stochastic path variation. Accordingly, context effects are attributed to tested search architectures, and the M4 claim is limited to path dependence under one schedule.

4.4 Directions for empirical validation

A useful empirical program would measure sequences rather than only cross-sectional associations. Repeated experience-sampling or micro-longitudinal classroom studies could record the visibility of an academic contribution, perceived peer response, subsequent participation, and context changes. The central M0 prediction is an interaction: a salient negative response should reduce later visible participation more when positive evidence is sparse. M1 predicts that lower-exposure participation

may preserve exploration, especially if students generalize safety across similar actions. M2 predicts that fresh-start opportunities should matter differently from simple reward changes. M3 predicts that clear signals about receptive contexts should reduce the context-count effect. M4 predicts that a history of behavior and peer response may matter even when current measured conditions are similar.

Such studies should preregister operational definitions, distinguish concealment of ability from privacy or strategic self-presentation, include non-gifted comparison groups where appropriate, and avoid treating gifted identification as a homogeneous exposure. Models could then be fitted hierarchically to individual sequences, compared against simpler learning rules, and evaluated out of sample. Ethical interpretation should preserve the possibility that selective disclosure is adaptive in some settings and that changing the environment, not the student, may be the relevant target.

4.5 Conclusion

Persistent masking is a robust learning outcome in these models, not merely a response to environments where masking is objectively best. Risky authenticity and hot-stove information restriction produced masking across 63.3% of the maintained M2 phase grid even though value iteration favored authenticity in those cells. Negative-outcome weighting and sticky stigma deepened avoidance; graded disclosure preserved safer exploration; multidimensional search made receptive contexts harder to discover; large comparison costs eliminated good-context authenticity; and endogenous receptivity made masking self-reinforcing and path-dependent. Together, the models show how a learner can become persistently hidden from opportunities that are, in objective terms, already worth taking. The later boundary conditions define what must be tested before translating these mechanisms into claims about prevalence or intervention in gifted students.

5 Data Availability Statement

The code, manifests, primary synthetic unit-level rows, summary tables, and verification scripts supporting this study are contained in the project repository. A permanent public archive and DOI must be inserted before submission: [AUTHOR: ADD ARCHIVED REPOSITORY DOI/URL]. No human participant data were generated or analyzed.

6 Ethics Statement

This study used computer simulations only and involved no human participants, human data, or animals. Ethical review and informed consent were therefore not applicable.

7 Author Contributions

[AUTHOR INITIALS]: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft, Writing - review and editing. [AUTHOR: revise this CRediT statement to match the final author list.]

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9 Conflict of Interest

[AUTHOR: Confirm before submission.] The author(s) declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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12 Figure Legends

Figure 1. Maintained directional contrasts across the model family. Points are differences between independent condition means; horizontal lines are descriptive 95% bootstrap intervals computed with 10,000 within-condition resamples. Positive values are in the expected direction stated on the y-axis. Intervals describe synthetic-unit dispersion within the recorded runs, not uncertainty across master seeds or a human population.

Figure 2. The large M2 regime of learned suboptimal masking. (A) Coarse phase map over receptivity p and stigma escape eta. Orange cells are the 226 of 357 cells (63.3%) classified as learned suboptimal masking: value iteration favored authenticity in the neutral state while the learned neutral action was masking. Gray cells were not assigned that classification. (B) Refined optimal neutral-state authenticity threshold under the default M2 environment. The dynamic threshold falls from 0.3839 to 0.2690 as eta increases and remains far below the static M0 threshold of 0.7167.

Figure 3. M3 action-space ablations. Each point is the K = 1 minus K = 8 difference in the proportion of learners whose modal final-window action was authenticity in the good context. Horizontal lines are descriptive independent-bootstrap 95% intervals. The flat and hierarchical formulations impose context search; the global-mask and fixed-good-context controls compress or remove it.

Figure 4. M4 ascending and descending continuation paths. The upper panel shows mean final-window mask rate and the lower panel shows mean final receptivity at each matched km value. The descending path inherits Q-values and receptivity from the ascending endpoint. The separation shows path dependence for this schedule; the seven points are serially dependent and are not independent replicates.

13 Tables

Table 1. Model family and principal findings.

Model	Decision structure	Mechanism isolated	Principal finding
M0	Mask vs. authentic bandit	Risky authenticity and hot-stove sampling	Risk and negative weighting sustained masking
M1	Five independent disclosure arms	Lower-exposure exploration	Graded disclosure preserved exploration
M2	Three states x two actions	Stigma, acceptance, turnover, and escape	Suboptimal masking occupied 226/357 grid cells
M3	Context and behavior choices	Search for a receptive peer context	Search architecture magnified context burden
M4	Two actions with evolving p_t	Behavior-environment feedback	Masking became self-reinforcing and path-dependent

Table 2. Maintained simulation contrasts.

Experiment	Conditions	Learners per condition	Steps / final window
M0 receptivity	p = 0.20 vs. 0.85	250	900 / last 20%
M0 negative weighting	lambda = 2.50 vs. 0.50 at p = 0.65	250	900 / last 20%
M1 receptivity	p = 0.85 vs. 0.20; five levels	250	900 / last 20%
M2 stigma escape	eta = 0 vs. 0.20 at p = 0.55	200	1,200 / last 20%
M3 context count	K = 1 vs. 8	220	1,200 / last 20%
M3 comparison cost	bf = 0 vs. 0.80; K = 4	220	1,200 / last 20%
M4 entrenchment	km = 0.003 vs. 0; p0 = 0.45	220	1,200 / last 20%
M4 ramp	seven km values, up then down	220 continued	500 per ramp point / last 20%

Table 3. Primary directional contrasts with descriptive independent-bootstrap intervals.

Contrast	Estimate	95% interval	n per condition
M0: lower masking at high p	0.733	0.721 to 0.745	250, 250
M0: more masking at high lambda	0.403	0.386 to 0.421	250, 250
M1: more disclosure at high p	0.569	0.556 to 0.582	250, 250
M2: less masking at high eta	0.201	0.115 to 0.287	200, 200

Contrast	Estimate	95% interval	n per condition
M3: lower good-context settling at K=8	0.200	0.141 to 0.264	220, 220
M3: lower good-context authenticity at high bf	0.918	0.882 to 0.955	220, 220
M4: more masking at high km	0.409	0.338 to 0.478	220, 220